

MATH 394 · PROBABILITY I — MINI PROJECT

The Birthday Paradox

How many people do you need in a room before it's more likely than not that two of them share a birthday?

23

people

is all it takes — for a **50.7%** chance

Introduction & Motivation

A simple question with a wildly unintuitive answer:

“How many people need to be in a room before it's more likely than not that two share a birthday?”

- There are 365 possible birthdays — most people guess an answer close to 183.
- The true answer is just 23 — far smaller than intuition suggests.
- This gap between intuition and math is why it's called a “paradox” — nothing is logically contradictory, it's just surprising.
- The same reasoning shows up in cryptography, hashing, and collision estimation.

MOST PEOPLE GUESS

~183

(half of 365)

THE REAL ANSWER

23

gives a 50.7% chance of a shared birthday

Setting Up the Model

Before computing anything, we fix simplifying assumptions.

1

365 equally likely days

Ignore Feb 29; every birthday is one of 365 days, each equally likely.

2

Independence

One person's birthday gives no information about anyone else's.

3

n people in the room

We want the probability that at least two of the n share a birthday.

GOAL

Find $P(\text{at least two of the } n \text{ people share a birthday})$

We'll call this event "match."

This is surprisingly hard to compute directly — next slide shows the trick.

Why Work With the Complement?

Computing “at least one match” directly is messy

A match could be any pair, or several pairs, or three people sharing one day, and so on — the direct count has many overlapping cases.

The complement is clean

“No match” means every single birthday is different — exactly one clean case to count.

$$P(\text{match}) = 1 - P(\text{no match})$$

Complement rule: an event and its complement always sum to 1.

Deriving $P(\text{no match})$

Assign birthdays to people one at a time, each one must dodge the days already taken:

Person 1

365 / 365

any day at all

Person 2

364 / 365

must avoid 1 taken day

Person 3

363 / 365

must avoid 2 taken days

⋮

⋮

⋮

Person n

$(365 - n + 1) / 365$

must avoid $n - 1$ taken days

Multiply these independent probabilities together to get $P(\text{no match})$.

The Formula

$$P(\text{no match}) = \frac{365 \times 364 \times 363 \times \dots \times (365 - n + 1)}{365^n}$$

Equivalently, as a product:

$$P(\text{no match}) = \prod_{k=0}^{n-1} (365 - k) / 365$$

$$P(\text{match}) = 1 - \prod_{k=0}^{n-1} (365 - k) / 365$$

Worked Example: $n = 23$

Plug $n = 23$ into the running product and watch $P(\text{no match})$ shrink:



$P(\text{no match})$

0.4927

$P(\text{match}) = 1 - P(\text{no match})$

50.7%

Why So High? Think in Pairs, Not People

It's tempting to compare 23 to 365 and expect a tiny probability. But a shared birthday can happen between any two people — so what matters is the number of pairs, not the number of people.

Number of pairs among n people:

$$C(n, 2) = n(n-1) / 2$$

For $n = 23$:

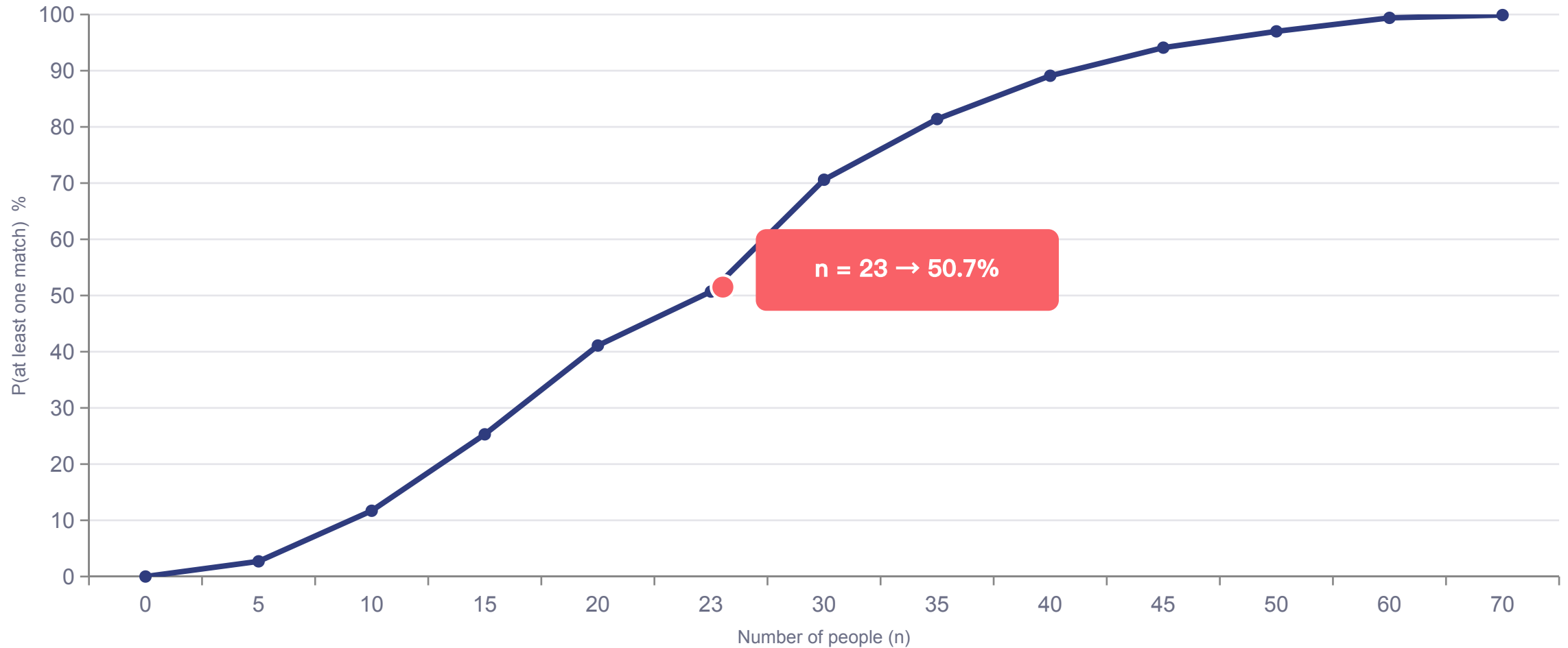
$$C(23, 2) = (23 \times 22) / 2 = 253 \text{ pairs}$$



Each of these 253 pairs is an independent “chance” for a coincidence — 253 chances adds up fast, even though each single pair is unlikely.

P(Match) Grows Fast, Then Flattens

Probability of a shared birthday vs. group size n



Application: Hash Collisions

The same “pairs, not people” logic drives collision estimates in computer science.

The setup

A hash function maps inputs to one of N possible outputs — just like birthdays map people to one of 365 days.

“ N ” plays the role of 365; hashed items play the role of people.

Birthday-attack estimate

Expected items before a collision $\approx 1.25 \times \sqrt{N}$

Far fewer than N — exactly the same surprise as $23 \ll 365$.

Concrete numbers

$N = 365$ (birthdays)

≈ 23 people for 50% odds

$N = 2^{32}$ (32-bit hash)

$\approx 77,000$ items for 50% odds

$N = 2^{64}$ (64-bit hash)

≈ 5.1 billion items for 50% odds

This is why cryptographic hash outputs must be long: short outputs collide far sooner than “ N guesses” intuition suggests.

Conclusion

Use the complement

$P(\text{match}) = 1 - P(\text{no match})$ turns a messy count into a clean product.

It's about pairs

23 people $\rightarrow$ 253 pairs — that's why the probability climbs so fast.

It's not just a puzzle

The same idea estimates hash collisions and other coincidence problems.

With just 23 people, a 365-day calendar isn't nearly as roomy as it feels.

Sources

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